Radiology AI | Implementation roadmap and test strategy

Frontend specification package - October 2026 | Planning baseline, not a repository audit

Milestones and quality gates

Phase 0 - discovery - Inspect backend repository, OpenAPI schemas, auth model, tenant model, existing AI endpoints and existing frontend work; document missing contracts and dependencies. Do not claim endpoints exist until verified.

Phase 1 - foundation - Create frontend repository, app shell, routing, localization, auth boundary, design tokens, API client, test setup, CI and synthetic mock server.

Phase 2 - clinical MVP - Build worklist, patient/study detail, report editor, clinician review/sign flow, loading/empty/error states, permission-aware navigation.

Phase 3 - specialized visualization - Integrate domain viewer and visualization with synthetic datasets; validate clinical usability and performance.

Phase 4 - AI workflow - Implement AI job submission/status, evidence presentation, accept/reject/edit actions and audit-facing feedback; no autonomous diagnosis.

Phase 5 - hardening - Accessibility and security review, threat modeling, browser compatibility, load testing, clinician usability sessions, localization review and production readiness checklist.

Definition of done - Every feature has acceptance criteria, type-safe contracts, test coverage, permission tests, no PHI leakage, clinical status labeling, responsive behavior and reviewer approval where required.

Feature-level tests

Workflow tests - Assigned worklist -> detail -> report draft -> review -> sign; authorization denied; stale version conflict; AI job timeout; interrupted upload; accessibility keyboard navigation.

Clinical visualization tests - DICOM viewer loads representative synthetic CT/X-ray studies and preserves study/series identifiers.

Security tests - No PHI in telemetry, error reports, browser storage, route params or public assets; no privileged UI action bypasses backend authorization.